Introduction

Let $f(x) = 2x$. Consider the following function:

$$F(x) = \int_0^x 2t \, dt.$$

1. Draw a graph of $f(x)$. On it, illustrate $F(1)$, $F(2)$ and $F(3)$.

2. What does $F(x)$ measure?

3. Without further computation, how do you know $F(x)$ is increasing? Concave up?

4. Calculate $F(1)$, $F(2)$, $F(3)$ and $F(-1)$ and draw a graph of $F(x)$.

5. What function do you think $F(x)$ is?

6. Use the FTC to prove your hypothesis from the previous question, then fill in the blanks below:

$$\frac{d}{dx}\int_0^x 2t\,dt = _____, \text{ so } \int_0^x 2t\,dt \text{ is an } _____ \text{of } f(x) = 2x.$$

7. Is $G(x) = \int_2^x 2t\,dx$ also an antiderivative of $f(x) = 2x$? If so, what constant do $F(x)$ and $G(x)$ differ by?

Stuff from the Past that will be Important Today!

⋆ **Extreme Value Theorem:** If $f(t)$ is continuous on the closed interval $[a, b]$, then it has a _____ _____ and a _____ _____ on that interval.

⋆⋆ **Bounding Integrals:** If $m \le f(t) \le M$ for $a \le t \le b$, then

$$\le \int_a^b f(t)dt \le$$

FTC II

The Second Fundamental Theorem of Calculus
Let f be continuous on an interval. Then for x and a in that interval

$$\frac{d}{dx}\int_a^x f(t)dt = f(x)$$

Proof: Suppose $f(t)$ is a continuous function and let $g(x) = \int_a^x f(t)dt$.

Then

$$\frac{g(x+h)-g(x)}{h} = \frac{\qquad\qquad\qquad\qquad}{h} = \frac{\qquad\qquad\qquad\qquad}{h}$$

By $(\star)$, since $f(t)$ is a continuous on $[x, x+h]$, then it attains a smallest value m and a largest value M. Then, by $(\star\star)$,

$$\leq \int_x^{x+h} f(t)dt \leq$$

Dividing everything by h, we get

$$\leq \frac{\int_x^{x+h} f(t)dt}{h} \leq$$

As $h \to 0$, what happens to the interval $[x, x+h]$?

As $h \to 0$, what happens to m and M?

Therefore, $\frac{d}{dx}\big[g(x)\big] = \lim_{h\to 0} \frac{\int_x^{x+h} f(t)\,dt}{h} =$ _____.

Questions

1. Why should you care (other than the test)? (Hint: can you write down an antiderivative for e^{-x^2}?)

2. Find the following derivative in two different ways: $\frac{d}{dx}\int_2^x \cos t dt$

   (a) Using FTC I:

   (b) Using FTC II:

3. Let $g(x) = \int_1^x \sqrt{1+t^2}\, dt$.

   (a) What is $g'(x)$?

   (b) What is $g(x^3)$?

   (c) What is $\frac{d}{dx}g(x^3)$? (Hint: you need the chain rule here. This is a composite function.)

4. (a) Find a function $g(x)$ such that $g'(x) = \sqrt{1+x^2}$ and $g(2) = 0$.

   (b) Find a function $g(x)$ such that $g'(x) = \sqrt{1+x^2}$ and $g(2) = 10$.

5. Let $g(x) = \int_0^x f(t)\,dt$, with $f(x)$ is continuous. Circle out the right answer for each of the following:

   - If $f(x) > 0$ and $x > 0$, then $g(x)$ is positive/negative and increasing/decreasing.
   - If $f(x) > 0$ and $x < 0$, then $g(x)$ is positive/negative and increasing/decreasing.
   - If $f(x) < 0$ and $x > 0$, then $g(x)$ is positive/negative and increasing/decreasing.
   - If $f(x) < 0$ and $x < 0$, then $g(x)$ is positive/negative and increasing/decreasing.

6. What constant do the following two antiderivatives of $f(x)$ differ by?

$$F(x) = \int_{-1}^x f(t)\,dt \qquad G(x) = \int_1^x f(t)\,dt$$

   (Hint: draw pictures!)